



Market Basket Analytics: Uncovering Purchasing Patterns



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Kick-off and Data Exploration

01



Overview of Domain & Analytics :

Financial Analytics: Risk assessment, fraud detection, investment and profitability.

Manufacturing Analytics: Predictive maintenance, quality control, production efficiency.

Human Resources (HR)

Analytics : Workforce planning, employee and performance.

Operations Analytics:

Streamlining supply chains, logistics, and process efficiency.

Healthcare Analytics: Patient outcomes, hospital resource allocation, predictive diagnostics.

Market Basket Analytics:

Behavior prediction and lifestyle modeling of customers.

Overview of Market Basket Analytics:



What is Market Basket Analysis?

Market Basket Analysis (MBA) is a data mining technique used to understand the purchase behavior of customers by analyzing the co-occurrence of items in transactions.



Why is it Important for Businesses?

MBA helps businesses identify product associations, optimize product placements, and enhance marketing strategies to increase sales and customer satisfaction.

Association Rules and Metrics of Market Basket Analysis:

Association Rules Explained



Association rules are expressions that describe how items are associated based on transaction data, providing insights that can guide decision-making in inventory and promotion strategies.

Support, Confidence, and Lift

Support measures the frequency of item sets in transactions, confidence indicates the likelihood of purchasing one item given another, and lift assesses the strength of a rule over random chance.



Common Primary Keys for Market Dataset :

Transaction ID :

A unique alphanumeric code assigned to a blockchain transaction, that allows reference to that specific transaction..

Items :

Specific products, that a buyer acquires during a transaction recorded in sales or procurement data.

Category :

Grouping of items that share common characteristics in order to organize them into logical and manageable groups..

Quantity :

The count or measure of goods in units, volume, weight or length bought being used to calculate total cost.

Price :

Amount of money a buyer is paying to acquire a product -the **exchange value** between the seller and the buyer.

Location :

The geographic place from where customers make purchases. It provides context for analyzing purchasing behavior.

Exploration of Raw Dataset Under Review (1) :

01

BILL_N°

04

PRICE

02

ITEM

05

PRESENT_DATE

03

QUANTITY

06

CUSTOMER_ID

07

COUNTRY



Exploration of Raw Dataset Under Review (2):

522,064

Total Number of Records :

2010–2011

Period of Dataset :

Not Available

Category of Items :

–9,600

Minimum quantity purchased :

80,995

Maximum quantity purchased :

1

Mode of quantity purchased:

10.09

Mean for quantity purchased :

3

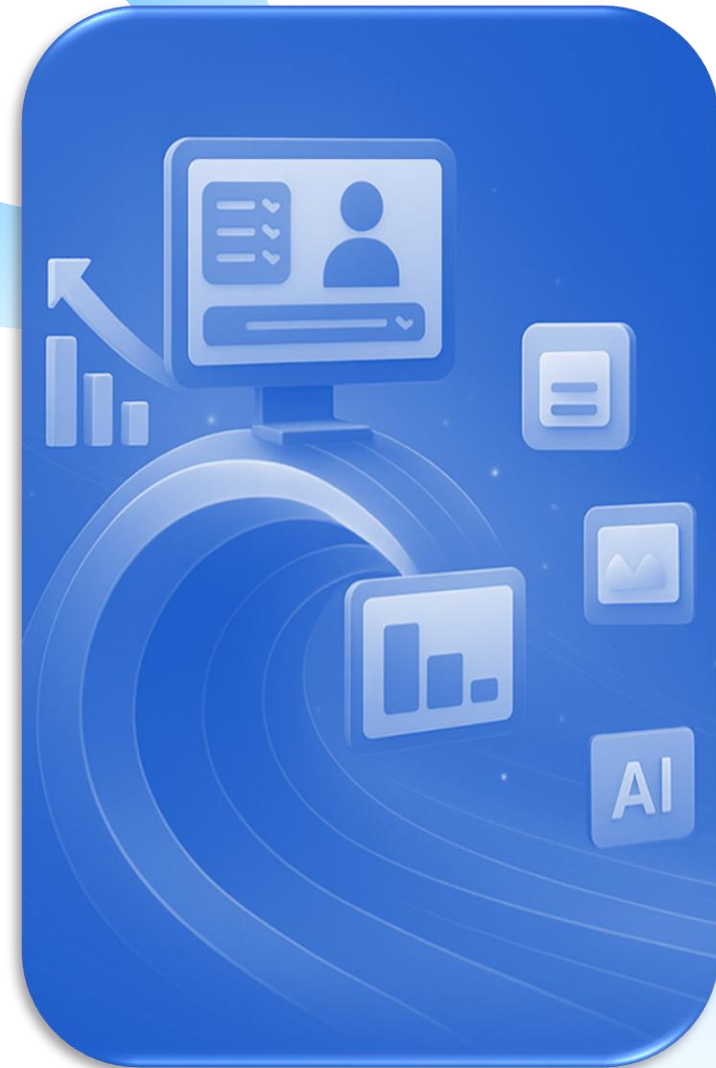
Median of quantity purchased :

Present

Outlier :

Data Cleaning and Transformation

02



Cleaning of the Dataset Under Review:

Data cleaning involves removing inconsistencies and handling missing values for data analysis.

01

Rows indicating “damaged”, “returned” or “posted items” were removed.

02

Rows indicating a negative value for quantity column were also removed.

03

Rows indicating a null value for item column were equally removed.

04

Rows indicating displaced values from other columns were corrected.

Transformation of the Dataset Under Review:

Data transformation involves converting data into a suitable format for analysis.

01

The dataset now has 517,735 records.

02

A new column for sales is now included by multiplying quantity with price.

03

Items were classified into top-level categories.

Top-Level Categories for Items:

01

ACCESSORIES

06

ORNAMENT

02

CARRIAGE

07

RECREATION

03

CLOTHING_TEXTILE

08

STATIONERY

04

CONSUMABLES

09

STORAGE_WARE

05

FOOD_WARE

10

YULETIDE_ITEMS

Data Visualization and Apriori Algorithm Implementation

03



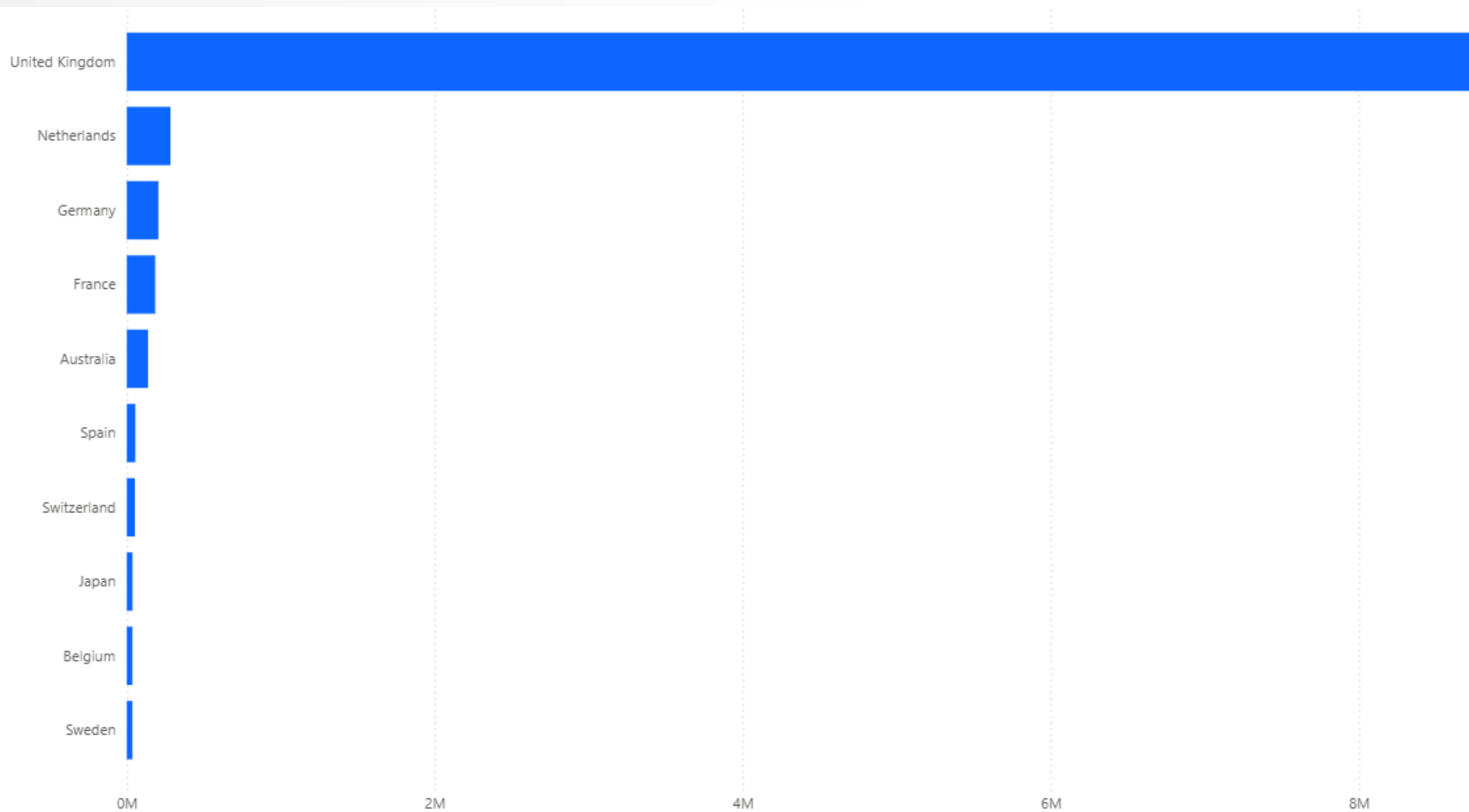
Sales by Country



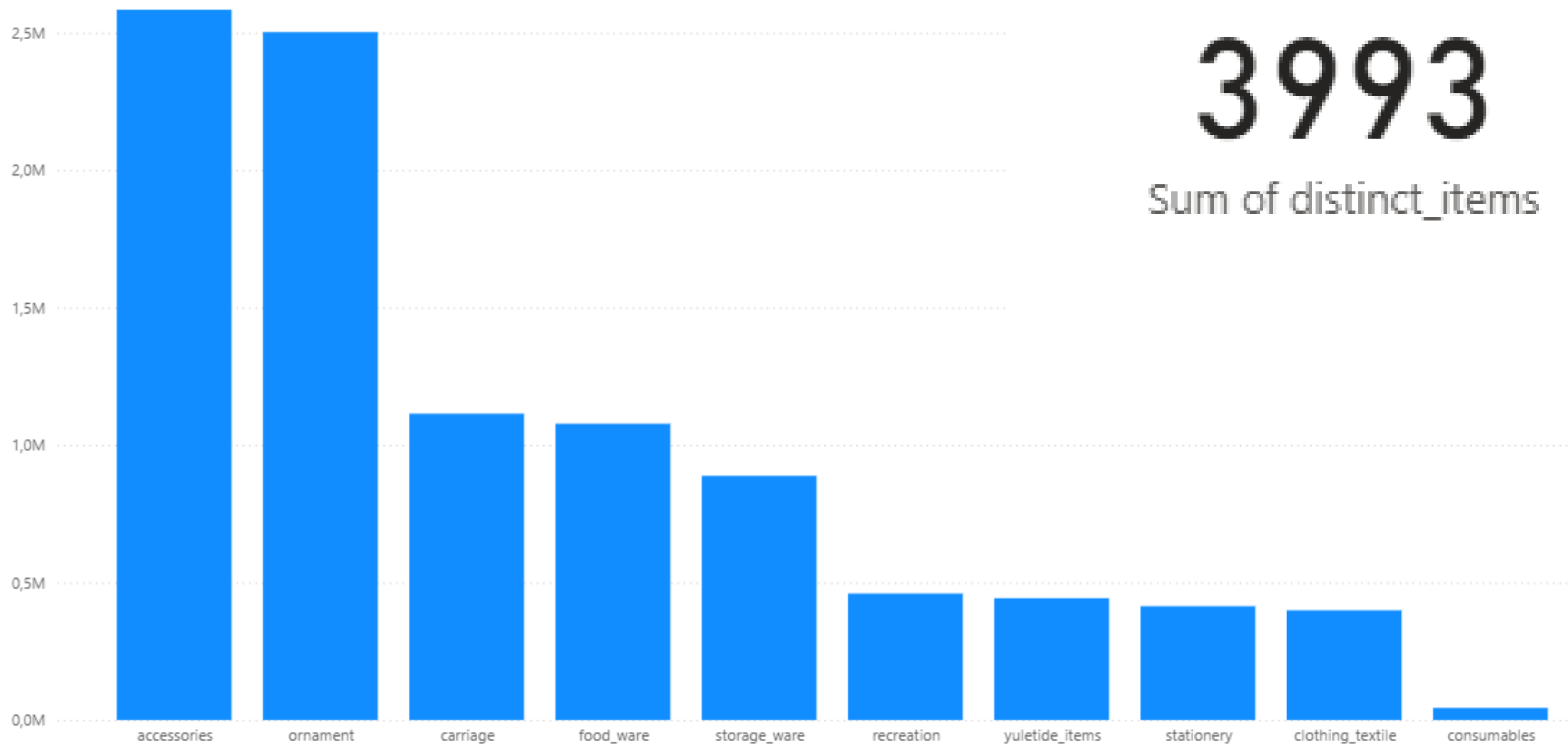
30

Count of Country

Top 10 Countries by sales



Sales by Category of Items



Implementation of Apriori Algorithm

Apriori Algorithm

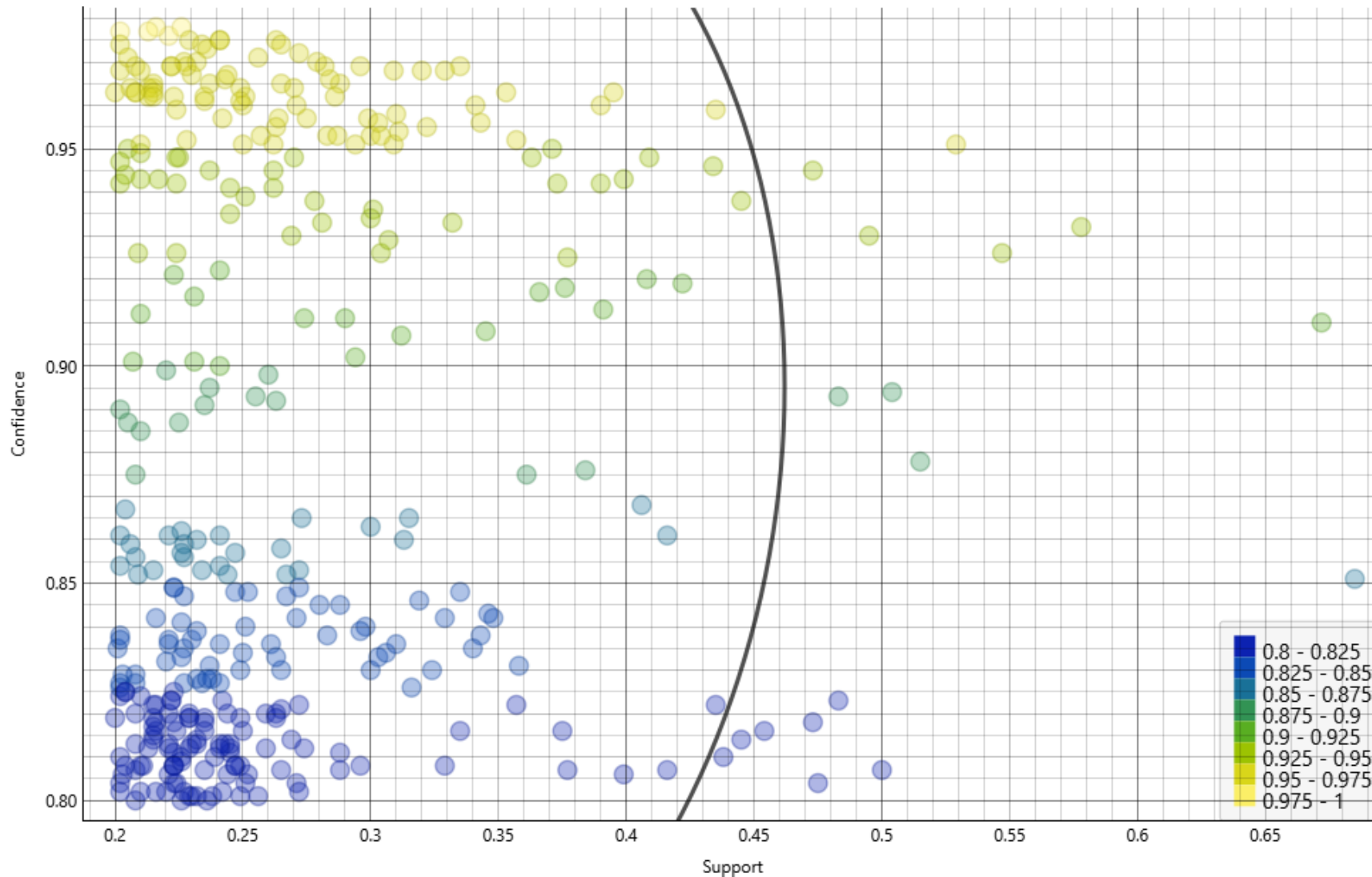
The Apriori Algorithm is a popular technique for mining frequent itemsets in transactional data. It uses a bottom-up approach to identify associations between items, allowing businesses to uncover valuable buying patterns.

Orange Data Mining

The dataset under review was converted to a binary matrix, where the arrangement of numbers in rows and columns were in two possible values: 0 or 1. This matrix was used as input for Orange Data Mining to detect association rules, which will be visualized with a scatter plot.

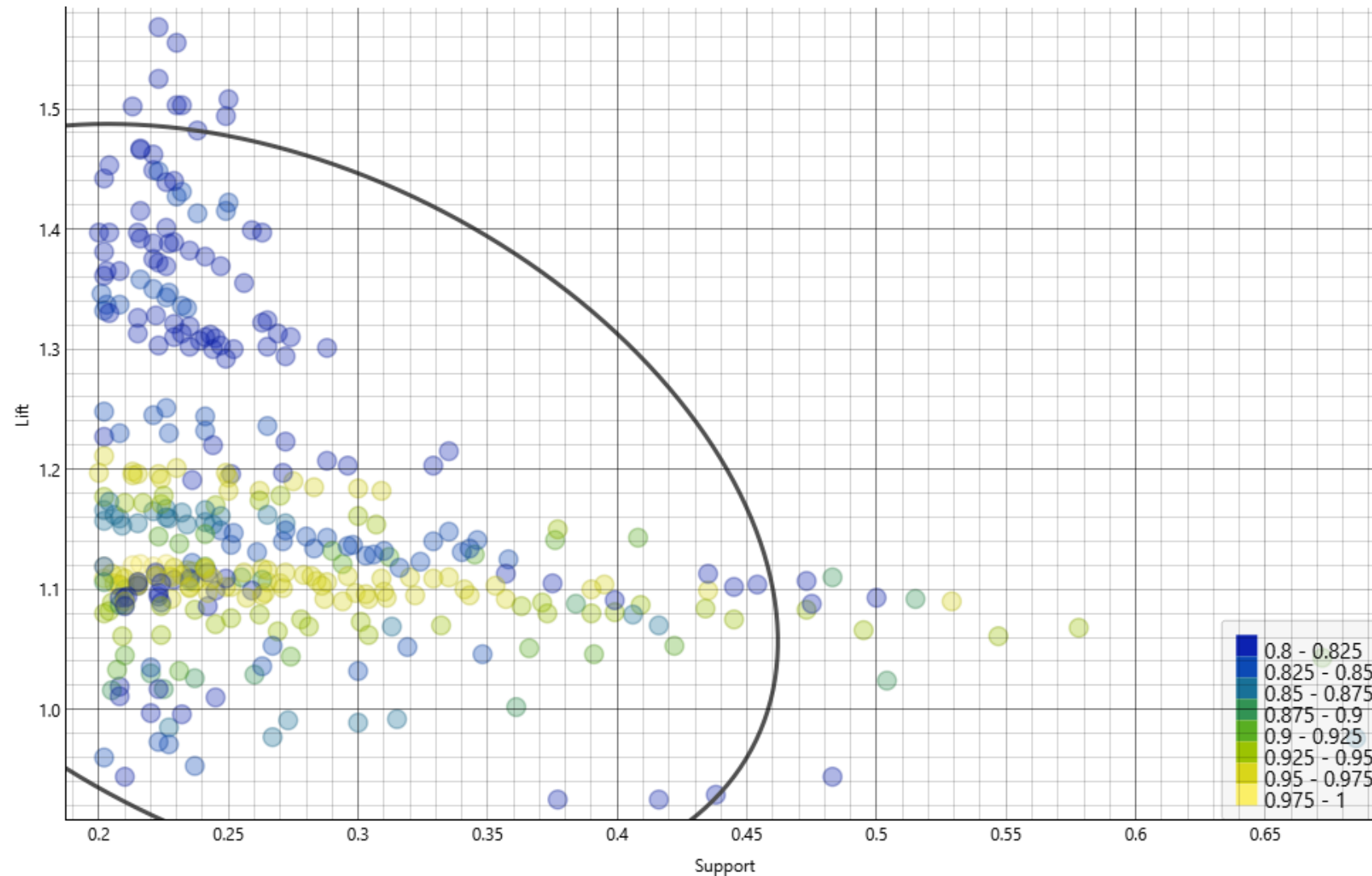
Scatter Plot for Association Rules (1):

Confidence: 80%, Support: 20%



Scatter Plot for Association Rules (2):

Lift: >1 , Support: 20%



**Insights, Recommendations,
and Finalization.**

04



Insights :

Insights (1):

Map & Bar Chart:

- ❑ There are customers from 30 countries across different continents, which indicates a global footprint, risk distribution, and revenue streams from various economies and cultures.
- ❑ A larger number of customers are from the United Kingdom, which indicates a strong dependency on the country.
- ❑ After the United Kingdom, there are also significant customers from other European countries, which implies that the orientation of Europe will greatly influence the demand cycles.

Insights (2):

Scatter Plot :

- ❑ Most of the sales were realized from the accessories and ornament categories of items.
 - ❑ There are 3993 distinct items classified into 10 categories.
- On the Scatter Plots, with **confidence at 80%** and **support at 20%**, there are more clusters at the left side of the chart.
- On the Scatter Plots, with **lift greater than one** and **support at 20%**, there are more clusters at the left side of the chart.

Finalization & Recommendation :



Finalizations :

- ❖ On Scatter Plots for association rules, good clusters are typically found in the upper-right region (high confidence and reasonable support) and highlighted by a lift that is greater than 1.
- ❖ The Market Basket Analysis does not suggest a strong association rule between categories of items presented in this analysis.

Recommendations :

- ❖ In a dataset of 3993 distinct items, 10 categories may not be enough to uncover the purchasing patterns of customers.
- ❖ A further analysis can be put in place to classify accessories and ornaments into sub-categories, which could help to uncover a strong purchasing pattern of customers.



Thanks for your audience